

Bryan Rhoelle Bergenholtz

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Education

California State University – Long Beach
Bachelor of Science in Computer Science

May 2025

Technical Skills

Languages: C++, Python, C#, TypeScript, JavaScript, SQL, HTML, CSS
Embedded Systems: ESP32, UART, Serial Communication, Firmware Development, JSON Parsing, Hardware Debugging
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Prisma, ASP.NET MVC, Razor, Vite
Databases: PostgreSQL, MySQL, MongoDB
Cloud & Tools: AWS S3, AWS RDS, GCP, Git, GitHub, Linux, Bash, Jira, Figma
Engineering Practices: Object-Oriented Programming, Data Structures, Algorithms, REST APIs, Code Reviews, Agile/Scrum, SDLC
AI Development Tools: GitHub Copilot, ChatGPT, Claude, Gemini

Experience

Terravan Solar June 2023 – May 2024
Software Engineer Intern

- Developed embedded software features for an ESP32-based battery monitoring system, enabling real-time diagnostic data exchange between embedded hardware and Android devices through structured JSON messages.
- Integrated and refined a C++ communication library for JBD battery management systems over UART, enabling access to voltage, current, temperature, capacity, and battery status metrics.
- Diagnosed firmware, serial communication, and hardware integration issues by inspecting device output, validating message formats, and testing communication behavior across system components.
- Collaborated with engineers during Agile sprints to implement firmware updates, document system behavior, troubleshoot defects, and improve long-term code maintainability.

Amazon Sortation Warehouse Nov 2020 – Aug 2021
Part-Time Associate

- Maintained workflow accuracy in a high-volume Amazon operation by interpreting digital sorting instructions, resolving scan errors, and escalating operational issues to team leads.
- Followed Amazon safety, equipment, and operational procedures while collaborating with associates and leadership under time-sensitive conditions.

Projects

Kinethic – Workout Tracking Web Application 2026 – Present

- Developed a workout tracking application using Next.js, React, and TypeScript for creating routines, recording exercise sets, managing rest timers, and reviewing workout history.
- Designed reusable components for workout sessions, exercise selection, user profiles, progression guidance, and same-day session completion states.
- Implemented structured application state and local persistence to preserve workout templates, completed sessions, profile settings, and training history across browser sessions.

DROCSID – CSULB Senior Capstone Project Jan 2024 – Dec 2024

- Built a real-time messaging application with authentication, file sharing, and REST APIs using React, Node.js, PostgreSQL, and Prisma.
- Designed persistent data models for users, messages, files, and application metadata while supporting efficient storage and retrieval.
- Collaborated in an Agile team through sprint planning, Git-based development, code reviews, debugging, technical discussions, and documentation using Jira and GitHub.

SRB Construction and Engineering Services Website June 2024 – Jan 2025

- Developed and deployed a responsive company website using Next.js and Tailwind CSS to improve access to the client's construction and engineering services.
- Gathered requirements through client meetings, created Figma prototypes, and refined the application through iterative feedback, testing, and design revisions.